EENG300013 Coursework Assessment

0 Introduction

0.1 Summary

The assessment for this unit comprises 3 parts:

Part 1. **Machines**: Converting system level requirements into machine requirements

Part 2. **Converters**: Analysis of the impact of DC link capacitance, filter inductance and PWM pattern on drive operation

Part 3. **Further design study** in one of the above themes (theme will be assigned to you).

The first two parts require you to replicate the analyses and simulations undertaken as part of the teaching in week 1 to 7 (each part covering one theme) but with a different set of parameters and conditions. For these parts (part 1 and 2) each student will receive a different parameter set and be required to submit a short structured deliverable for each part, which is detailed later in this document. Parameter sets will be issued in week 08.

Part 3 requires you to use the parameters, analyses and simulations you have developed for the case study during weeks 1 to 7 and undertake a more in-depth engineering design analysis in a theme that will be given to you by the teaching team. Within the theme that you are allocated (machines or converters) you are free to choose the design study you undertake with some examples being given later in this document

0.2 Submission

For each part you will be required to submit at least 2 files. Details on the submission point will be released in week 08.

1. The deliverable documents for that part (detailed for each part separately in this document)
2. A single working simulation that demonstrates the simulation(s) that you have undertaken for that part to get the outcomes you derived.
3. OPTIONAL – m-file used to calculate the values you use in your simulation for it to run.

0.3 Deadline

The single standard deadline for all 3 parts is 13:00 on Thursday of week 11 (3rd December 2026).

Deadline extensions need to be requested formally via the school using the University's standard process and should be requested at least 48 hours in advance of the deadline in question. These requests will normally require supporting evidence.

0.4 Plagiarism & Cheating (extracts copied from [2]): The rules

The coursework is to be undertaken assuming that the use of AI tools fall under the **Category 2 minimal** usage scenario as describe here:

https://www.bristol.ac.uk/bilt/sharing-practice/guides/guidance-on-ai/using-ai-in-assessment/

This states that:

*"for example, using spell & grammar checkers to help identify mistakes but **not** rewrite chunks of text.* **"**

The coursework assessment from EENG30013 in 2026/27 is **individual submission** and **open-book**. According to university regulations, this means:

a) Use of resources is allowed provided these are referenced appropriately. Insufficient referencing constitutes plagiarism and typically leads to an academic misconduct hearing, and penalty.

b) No external help can be used. Use of external help is considered cheating, and leads to a formal investigation.

In other words, everything that you submit should represent your own ideas and work, with anything else being appropriately described and referenced.

0.4.1 Post Assessment Vivas

In order to provide a robust approach to ensuring students have met the intended learning outcomes of the unit, and to encourage the appropriate and professional use of Artificial Intelligence-based tools, a random sample of assessment submissions will be selected for post-assessment in-person vivas.

These vivas will take place after the assessment submission and will require students to attend an in-person interview so they can discuss their submission with members of the unit team. The interview panel, consisting of two academic members of staff, will ask questions to determine the level of familiarity with and depth of understanding of the submission, and the materials and processes used to create it.

0.4.2 What to expect

A post-assessment viva is nothing to worry about. You simply need to talk through the work that you have done and it's great practice for future interviews or project vivas in your third or fourth year. You don't need to bring anything with you, but you may find it beneficial to bring your laptop, or any notes you used to create your assessment, to help you explain your process.

If the post-assessment viva panel detects any irregularities or academic misconduct after academic review, the case is referred to the appropriate procedure for investigation and/or consideration, as set out in the University's Academic Misconduct Procedure. To ensure you are meeting all University guidelines, we recommend you read your assessment brief fully and ensure you remind yourself of the University's policy on academic integrity here: Academic Integrity

Make sure you follow all assessment brief guidance fully and familiarise yourself with academic integrity.

0.4.3 Avoiding plagiarism

Use citations (or references) and a bibliography to state clearly ALL resources that you have used, including:

- teaching materials not in this unit,
- ideas from websites,
- books,
- papers,
- other people, etc

You must be transparent in your writing how these resources have inspired your work. This means that it should be clear about whether you are including other people's ideas, arguments, structure, literature reviews , methods, graphs, figures, logic, code, assumptions, or maths. For example, it would be plagiarism to use a review paper to help you find useful papers, without explicitly saying how this review paper helped you do so. This would be correct: "In [x], a research team from… have reviewed a range of heatsink designs, and the most efficient being water-cooled [x], phase change [x], and …"

Therefore, for your referencing to be effective, you must

- state where the influence came from using a reference, e.g. "[4]" plus an entry in your bibliography, AND
- make clear what you used from this reference, e.g. "in [4], a model is introduced to explain why the conductivity of materials is a function of temperature. In this work, this model was combined with an expression found in [5], that models the dependency on humidity, to create the model below:"

In other words, placing a "[x]" will not protect you from plagiarising.

0.4.4 How to avoid cheating

Some students cheat without knowing they are cheating. The following is to help you understand the university rules on this, so that you don't get investigated for cheating. Here are some traps that some students might fall into because they mistakenly think that these are not cheating:

- Writing in your mother tongue, and uploading paragraphs to an online translator. Whether this is an algorithm, artificial intelligence, or a person who is being paid, the resulting work is not your work, and this practice is classified as cheating.
- Having an artificial-intelligence-based tool proofread your English is cheating, since it will paraphrase and reword, and do work that an assessor might think you did.
- Using a paraphrasing tool to reword a paper, and then referencing this paper. This is both plagiarism and cheating at once! It is plagiarism, because you are using ideas, structure, thought processes, argumentation of someone else; and your reference "[x]" has not made it clear what exactly you are using from this paper. It is also cheating as the paraphrasing tool is carrying out writing work for you.
- Using an artificial-intelligence-based tool to draft a first version of an assignment that you then edit is cheating. The only way of doing this without being done for academic misconduct, would be to explain your method, e.g. "There are many types of humidity sensor, in fact on prompting an AI tool to write a list of these using the prompt "list all types of humidity sensor", the following list was compiled…. The AI tool was then prompted to categorise these by physical operating principle, using the prompt "…", upon which it listed …". Since full transparency was applied here, you would not get done for cheating. However, ask yourself if it is good engineering practice to work in this way. It the AI tool tested by experts to produce reliable work in this area? Have you validated all the findings? Is this useful for the reader? Are you demonstrating the skills that the assessor is looking for? It would have been better to look for reliable (rigorously expert-reviewed) review papers or to attempt this task yourself.

We will use the University's standard plagiarism and online tool checking processes to ensure that any plagiarism/cheating is fairly penalised in accordance with the University's regulations. If you do not follow this advice and we find material that is duplicated online or in other students' work we will treat the case as a case of cheating /plagiarism and refer it to the Academic Integrity Team for consideration.

0.5 Assessment Criteria

All 3 parts of the assessment will be assessed and feedback given using the assessment criteria rubric in Figure 0-1.

- Each of parts 1 and 2 will be graded up to a 2:1 in ILOs 1 and 2 and allow a pass/fail in ILOS 3 and 4
- Part 3 will allow pass/fail or improvement of ILOs 1 and 2 and be graded in ILOs 3 and 4.

Based upon your performance overall in all 3 parts of the assessment a decision on pass/fail for the unit will be determined (requires a pass in each ILO) and a single grade determined based upon your performance overall.

0.6 Bibliography

[1] "14. Marking Criteria and Scales" | Regulations and Code of Practice for Taught Programmes https://www.bristol.ac.uk/academic-quality/assessment/regulations-and-code-of-practice-for-taught-programmes/marking-criteria/, (Accessed 21st March 2023)

[2] Extracts taken directly from - Stark, B. – "A guide to writing engineering reports" – *Internal support document* – Department of Electrical and Electronic Engineering, University of Bristol, 2023

Unit ILO	Contributing assessment parts	Fail						3rd Class			2:2 (Lower Second) Class			2:1 (Lower Second) Class			First Class			Exceptional First Class		
		0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
		0	7	15	22	29	35	42	45	48	52	55	58	62	65	68	72	75	78	83	94	100
ILO 1: Describe and select the fundamental elements of an electric drive system and the benefits that these elements bring to a particular application.	Parts 1 and 2	No/unsufficient evidence of understanding of drive technologies or how these interface with one another						Completes at least some of the requirements with some errors replicating some of the design studies taught in the lab exercises for at least 1 of the 2 elements of the drive (machine/converter)			Completes many of the requirements correctlty of the design problem for at least 1 elements of the drive (system/ converter)			Completes all of the requirements correctly for at least 1 element of the design problem (machine/ component)			Completes all of the requirements correctly for at least 2 element of the design problem (machine/ component)			Part 3 of the assessment provides evidence to access these criteria below		Completes all of the requirements correctly for all elements of the design problem and extends these analyses proposing new context-appropriate solutions/ideas
ILO 2: Source, extract and use appropriate data to populate relevant models for design evaluation.	Parts 1 and 2	Significant or hazardous evidence of incorrect use of datasheets/ design data for problem analysis/ simulation						Identifies data for relevant components to populate simulation and/or uses relevant data from design problem in simulation/ analysis for at least 1 element of the drive			Extracts simple data from appropriate data sheets/design problem data and uses these in an analytical design problem or simulation correctly in most cases for at least 1 element of the drive.			Extracts data from appropriate data sheets/design problem data and uses these in an analytical design problem or simulation appropriately in all cases for at least 1 element of the drive .			Extracts data from appropriate data sheets/design problem data and uses these in an analytical design problem or simulation appropriately in all cases for 2 elements of the drive .			Part 3 of the assessment provides evidence to access these criteria below		Extracts new data from externally sourced data sheets/design problem data and exploits their benefits in design problem using developed tools to quantify and demosntrate confidence in their ideas
ILO 3: Design and evaluate an electric drive based upon an appropriate converter and machine topology for the various elements of the system using a mixture of analytical techniques and simulation tools across multiple domains for example: electrical, mechanical and thermal.	Parts 1, 2 and 3	Little evidence or incorrect application of appropriate analysis/simulation method resulting in no or very poorly formed outcomes.						Undertakes an analytical method and simulation for the design problem. Selects an appropriate simulation method for the task being undertaken for each task. Builds and uses simulations to test outcomes appropriately and provide cross validation to build confidence in design tools.			Part 3 of the assessment provides evidence to access these criteria below											
											Uses developed design tools (analytical framewortk and simulation) to explore additional operating points and come to simple conclusions.			As 2:2 with clear evidence and reporting of more context aware exploration in at least one element of the design problem (system, converter) as it is presented. Findings all appropriate.			Demonstrates a clear line of thought in producing a solution to the design problem. As for 2:1 but with a clear demonstration of exploring the design problem with creative selection/estimation of design deviations for further exploration or with a clear design intent and reflection.			Providing an analysis that undertakes a thorough context-aware design analysis over multiple/all elements of the problem and extends beyond the original design specification and in the context of the application/ system problem defintion and across multiple domains.
ILO 4: Reflect on the strengths and weaknesses of a design solution with respect to the application requirements using appropriate evaluaton metrics.	Parts 1, 2 and 3	No/Poor documentation of and reflection on design process and outcomes.						Uses results to detemine engineering requirements that satisfies a the simple required design tasks with little reflection			Part 3 of the assessment provides evidence to access these criteria below											
											Produces results that satisfy many of the required design tasks as presented but at additional operating points with limited reflection.			Produces results that satisfy most of the required design tasks with discussion of trade-offs required and/or simplifications/ assumptions made, or details added and reflection on the process.			Clear reporting of the design path and outcomes and indentification of path for further exploration were there more time and/or clear reflection on the areas of uncertainty in the results.			Clear evidence and concise reporting of well thought out multi-domain design exploration and refelction on the outcomes in the context of the system /application rather than just the technology.

Figure 0-1: Assessment criteria for EENG30013

1 Part 1 - Machines

1.1 Assessment Requirements – Machines

You are required to analyse the mechanical drivetrain of an elevator mechanism (

Figure 1-1) based upon the following elements

- a permanent magnet synchronous motor and a DC link voltage value as an input
- **intermediate twin-ratio gearbox – different from case study**
- counterweight, lift car and load suspended/driven by rope-drum.

The form of the drivetrain is the similar to the one used in the case study in activity sheets but contains an extra gearbox. Note also that you will be analysing a drivetrain with different parameters to the one used in the case study in weeks 01 to 03.

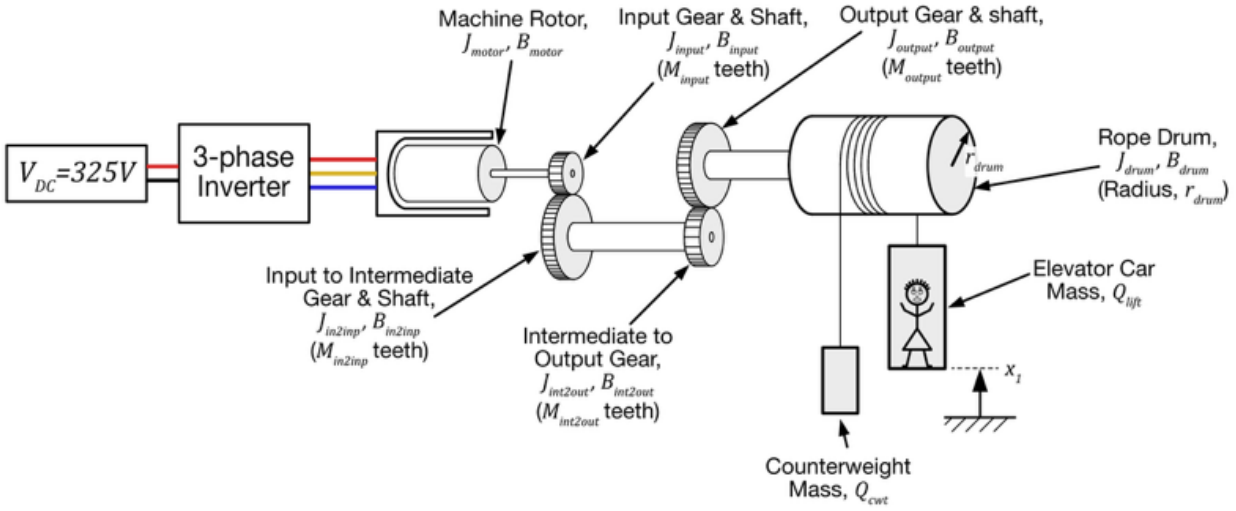


Figure 1-1: Elevator Drivetrain Schematic

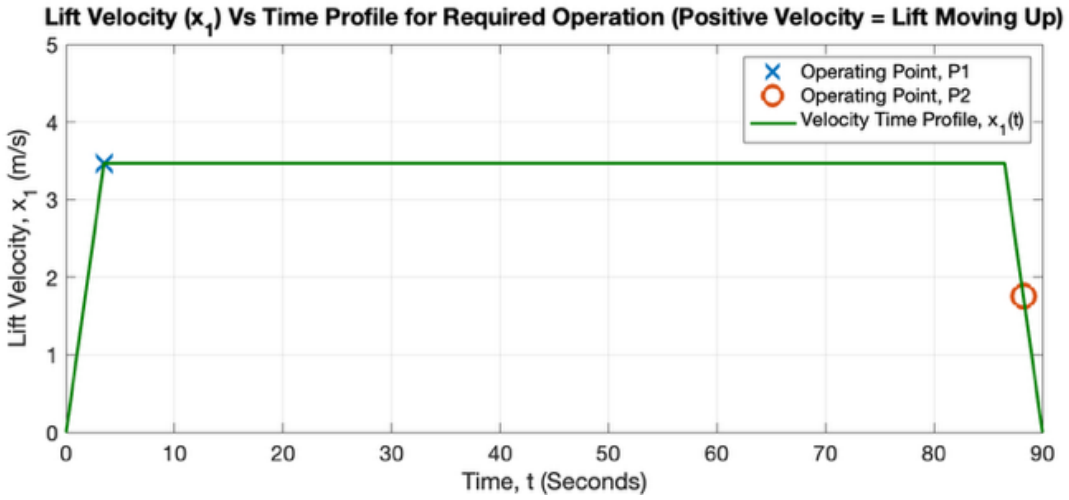


Figure 1-2: Example Speed vs Time Curve (or high level specification)

You will also be provided with a high-level specification (speed vs time curve) that the lift needs to be able to achieve at the output. An example based upon the case study is shown in Figure 1-2.

The parameters of the drivetrain, motor, load and the high-level specification will be provided towards the end of week 08 through blackboard. There are a range of parameter sets, you will be required to undertake the tasks listed below based upon the parameters set you are given. Note, if you use an alternative parameter set to the one you have been assigned you will not pass this aspect of the assessment.

Based upon the high-level specification provided, the task is for you to determine the

(i) Motor angular velocity Vs time profile
(ii) Motor electro-magnetic torque Vs time profile
(iii) Current Vs time profile (d and q axis and total current magnitude)
(iv) Voltage Vs time profile (d and q axis and total voltage magnitude)

and present these using four annotated waveforms similar to those shown in Figure 1-3.

You must also justify the appropriateness of the motor to perform the required loaded speed-time profile based upon the motor specification and the available DC link (3 sentences).

ILOs 1 and 2 marked against:

- correctness of waveforms
- correctness of annotation values for parameter combination
- waveforms (axes labels, white background, >5 cycles etc.),
- clear annotations.

ILO 3 pass awarded for suitable 100-200 word statement that demonstrates clearly how you have derived confidence in your results (i.e. simulations, multiple methods, etc) and elaborates any results you are less confident in and why.

ILO 4 pass awarded for determining correctly if the DC link suggested in the problem could be used successfully with the motor and selected drivetrain and why.

The solution you derived should be uploaded to blackboard along with a simulation that you have used to derive results.

1.2 Example Parameter Set (DO NOT USE THESE)

1.2.1 Lift requirements

The lift must be able to travel a distance of 300m in 90s.

For the purposes of this exercise, you are asked that the speed vs time profile have the form shown below:

- accelerate the lift linearly to the steady-state velocity, $\dot{x}_1^{steady} = 3.5m/s$, from $t = 0s$ to $t = 3.5s$,
- maintain the lift speed at steady-state velocity, $\dot{x}_1^{steady} = 3.5m/s$ from $t = 3.5s$ to $t = 86.5s$,
- decelerate the lift linearly from the steady-state velocity, $\dot{x}_1^{steady} = 3.5m/s$ from $t = 86.5s$ to $t = 90s$.

This results in a linear velocity Vs time profile $x_1(t)$ for the lift that has the form shown in Figure 1-2.

1.2.2 Example Elevator Drivetrain (Load) Parameters

An example of the parameters that describe the details of the load shown schematically in Figure 1-1 are given in Table 1-1. You will be given a set of these for your coursework,

Table 1-1: Mechanical Load Parameters:

Parameter		Symbol	Value	Units
Linear Elevator System	Mass of lift car	Q_{lift}	600	kg
	Mass of "average" person	Q_{person}	80	kg
	Rated capacity of lift	N_{person}	10	people
	Counterweight mass	Q_{cwt}	100% of lift mass + 40% of rated load mass	kg
Rope Drum	Radius	r_{drum}	0.2	m
	Mass moment of inertia	J_{drum}	9.912	kg m^2
	Viscous friction in drum	B_{drum}	10	Nm s rad^{-1}
Output Gear	Mass moment of inertia of gear and shaft	J_{output}	1	kg m^2
	Viscous friction of gear and shaft	B_{output}	0.01	Nm s rad^{-1}
	Number of teeth	M_{output}	60	teeth
Intermediate to Output Gear	Mass moment of inertia of gear	$J_{int2out}$	0.5	kg m^2
	Viscous friction of gear	$B_{int2out}$	0.05	Nm s rad^{-1}
	Number of teeth	$M_{int2out}$	30	teeth
Input to Intermediate Gear	Mass moment of inertia of gear and shaft	J_{in2int}	0.06	kg m^2
	Viscous friction of gear and shaft	B_{in2int}	0.002	Nm s rad^{-1}
	Number of teeth	M_{in2int}	30	teeth
Input Gear	Mass moment of inertia of gear and shaft	J_{input}	0.0123	kg m^2
	Viscous friction of gear and shaft	B_{input}	0.001	Nm s rad^{-1}
	Number of teeth	M_{input}	20	teeth
Input	Supply DC Link Value	V_{DC}	325.3	V_{DC}

1.3 Machine Parameters

An example of the parameters that define the 3-phase, AC, Permanent Magnet Synchronous Machine (PMSM) machine are given in **Table 1-2**. You will be given a set of these for the coursework.

Table 1-2: Machine Parameters

Parameter		Symbol	Value	Units
MACHINE TYPE: 3-phase, AC, Permanent Magnet Synchronous Machine				
Electrical	Stator phase resistance	R_s	0.054	Ω
	Stator d-axis synchronous inductance	L_{ds}	0.00285	H
	Stator q-axis synchronous inductance	L_{qs}	0.00285	H
	Permanent Magnet Flux	λ_{PM}	0.8603	Wb
	Pole Pairs	P	3	
Mechanical	Rated machine rotor rotational speed	ω_R^{rated}	1000	rev/min
	Maximum machine rotor rotational speed	ω_R^{max}	1500	rev/min
	Maximum machine rotor angular acceleration	α_R^{max}	25	rad/s^2
	Mass moment of inertia of machine rotor and shaft	J^{total}	0.25	kg m^2
	Viscous friction of machine rotor and shaft	B^{total}	0.005	Nm s

1.4 Example solution - Machines:

1.4.1 Key attributes (Things we are looking for when determining a mark)

1. Angular Velocity waveform – Annotation labels maximum angular velocity and states angular acceleration at this point
2. Torque speed waveform – Annotation states and highlights the maximum torque requirements AND the mechanical power at this point
3. Current waveforms – Annotation states and highlights the maximum total current magnitude requirements AND the electrical power at this point
4. Voltage waveforms – Annotation states and highlights the maximum total voltage magnitude requirements AND the electrical power at this point
5. Annotate either the voltage OR current graph showing the point at which maximum electrical power is calculation if this is different to either or the points in 3 or 4 above.
6. Correct statement and values chosen from below concerning suitability of DC link
    a. Motor can undertake the task based upon the available DC link and meets all specification requirements.
    b. Motor can undertake the task but NOT based upon the available DC link – Suggest DC link level that would be required
    c. Motor cannot undertake the task based as at some point in the speed time profile the motor maximum ratings are exceeded.
7. Statement (around 100 words) clearly communicating how you have derived confidence in your final results. Try and write this as if you were trying to convince a chief engineer that need to approve or sign off your results.

1.4.2 Demonstration – Machines

1.4.2.1 Analysis/Simulation Outcomes (ILOs 1 & 2)

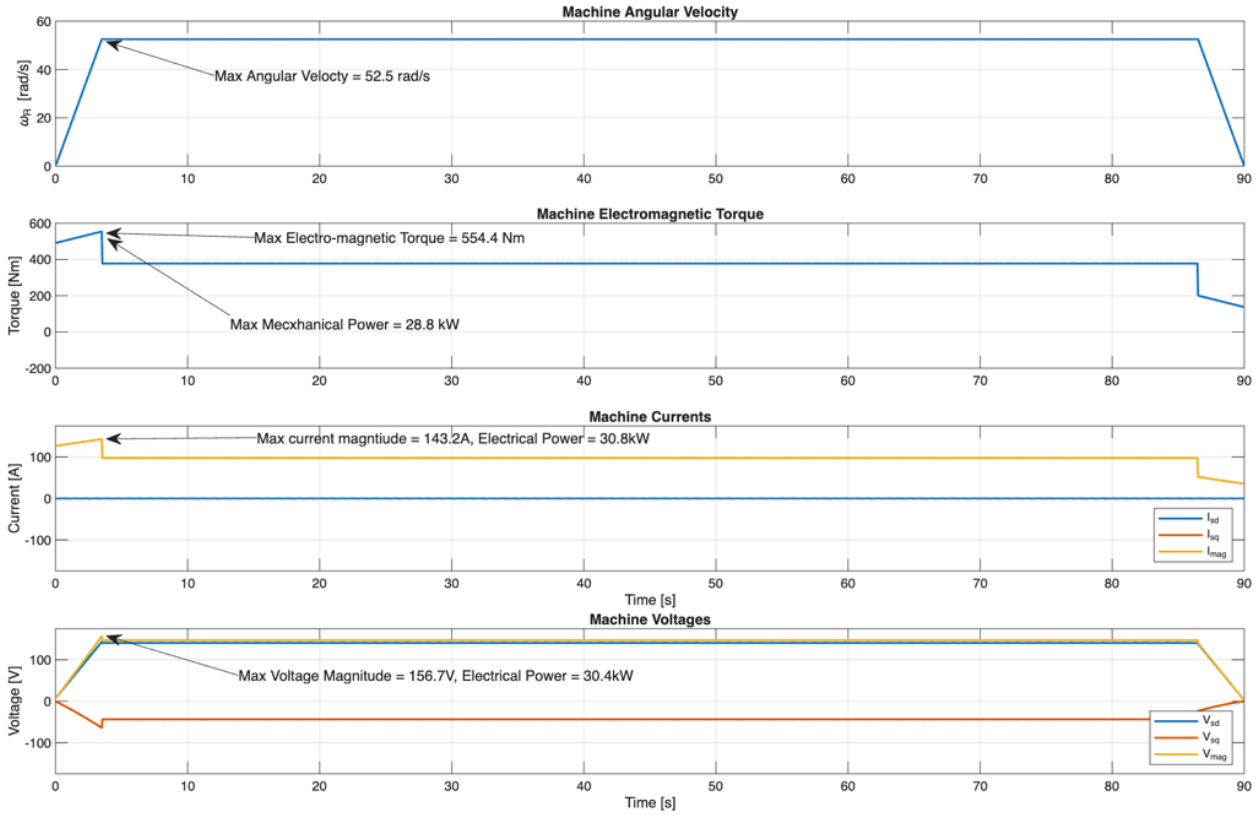


Figure 1-3: Example Figure Submission for Part 1 (not the actual answer)

1.4.2.2 DC Link Suitability (ILO 3 – Pass/Fail)

DC link voltage = XXXV and maximum motor voltage magnitude is XXXV hence the motor and DC link are suitable for the task.

1.4.2.3 Communicate How You Have Derived Confidence in your Results (ILO 4 – Pass/Fail)

I derived confidence in my results using the follow technique(s)/approach(es) … .

I'm less certain in the following results because….

2 Part 2 – Converters

2.1 Assessment Requirements – Converters

You are an electrical engineer working as part of a power electronics and drives consultancy. A customer has commissioned a project aimed at identifying a suitable DC-link capacitor for a three-phase motor drive they are designing for a large industrial fan. They have down-selected two candidate capacitors, shown in Table 2-1, which you are required to evaluate and subsequently recommend the component that is most appropriate for their system. Table 2-2 shows a list of operating parameters that have been provided to you by the company for their system.

Provide a brief report that includes:

- For the selected component, four annotated waveforms showing: *(ILO1 and ILO2)*
  1. DC-link inductor current Vs time profile
  2. DC-link inductor voltage Vs time profile
  3. Capacitor current Vs time profile
  4. Capacitor Voltage Vs time profile
- A statement on how you derived confidence in the final result *(ILO3)* – up to 100 words
- A judgement on the design task *(ILO4)* – up to 50 words

Table 2-1: Down-selected capacitor options

Capacitor option 1:		Capacitor option 2:	
Capacitance		Capacitance	
ESR		ESR	
Ripple current rating		Ripple current rating	

Table 2-2: Drive operating parameters

Parameter	Value
Vd	-44.7
Vq	105.2
Iphpk	110A (Id=0A)
Switching frequency	10000Hz
Modulation Scheme	SPWM
Maximum Vdc ripple	3%
Supply voltage	325V
Fundamental Frequency (f1)	40Hz

2.2 Example solution - Converters:

2.2.1 Key attributes (Things we are looking for when determining a mark)

1. L_{dc} current waveform with an appropriate annotation
2. L_{dc} voltage with an appropriate annotation
3. Capacitor current Waveform – Annotation shows I_{ripple}
4. Capacitor voltage waveform – Annotation shows V_{DC} ripple ($V_{esr} + V_c$)
5. Correct statement outlining the suitability of the two DC-link capacitor options (50 words).
6. Statement (around 100-200 words) clearly communicating how you have derived confidence in your final results and elaborates any results you are less confident in and why. Try and write this as if you were trying to convince a chief engineer that need to approve or sign off your results.

2.2.2 Demonstration – Converters

2.2.2.1 Analysis/Simulation Outcomes (ILOs 1 & 2)

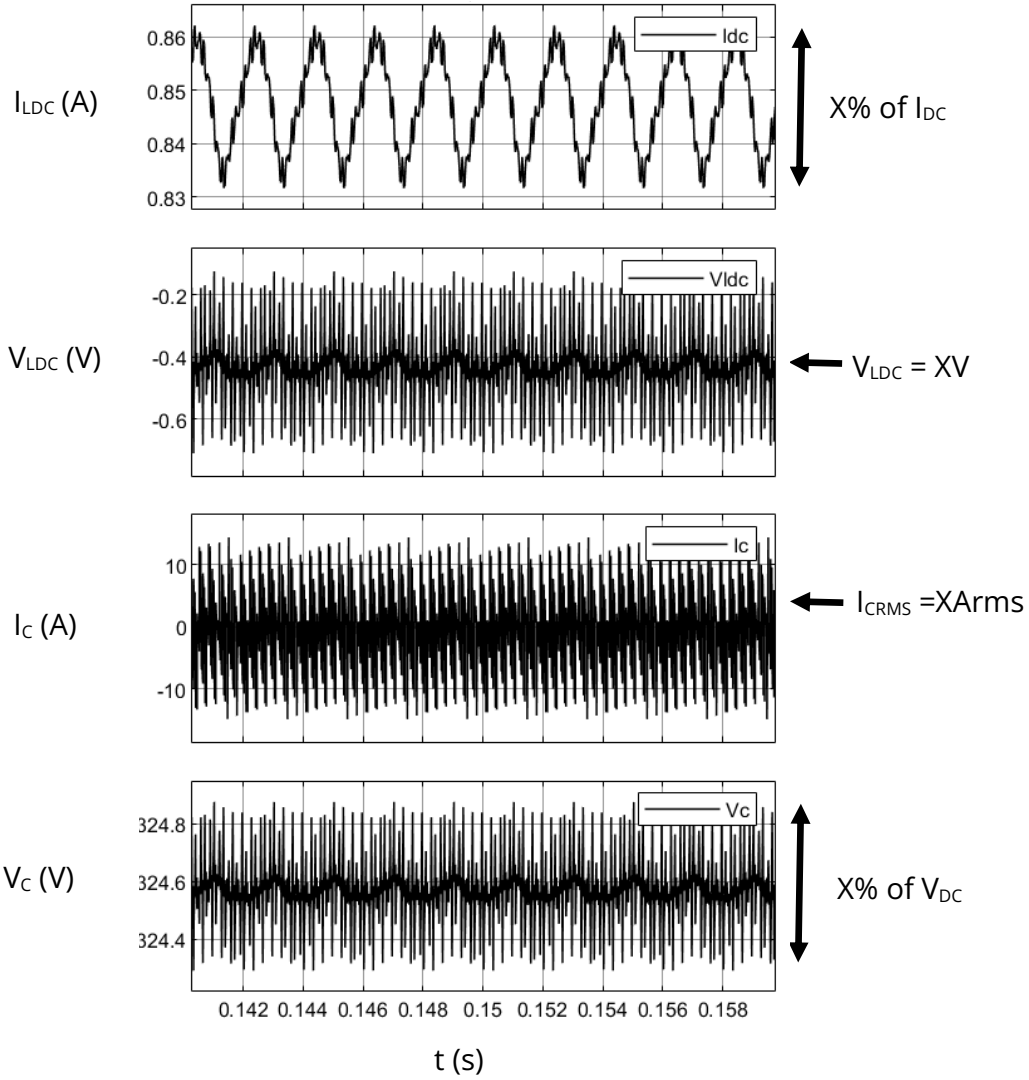


Figure 2-1: Example Figure Submission for Part 2 (not the actual answer)

2.2.2.2 DC Link Suitability (ILO 3 – Pass/Fail)

Capacitor option X parameters meets the requirements whereas Capacitor option Y does not due to….

2.2.2.3 Communicate How You Have Derived Confidence in your Results (ILO 4 – Pass/Fail)

I derived confidence in my results using the follow technique(s)/approach(es) … .

I'm less certain in the following results because….

3 Part 3 – Further Study

3.1 Assessment Requirements – Further Study

For part 3 you will be assigned a theme from one of the following:

1. Machines
2. Converters

Using the **case study parameters from weeks 1 to 7** (not the ones you have been assigned for the assessment) and the analyses/simulations you have developed during the teaching in weeks 1 to 7, undertake a design study that explores an aspect of your assigned theme in more detail.

Alternatively, you could use the theory presented in the Switches lectures in week 7 to develop an analysis and simulation to determine the loss in the inverter switches over a single AC cycle at the case study maximum speed.

Imagine that you are attempting to convince the technical director of your future company of the suitability (or not) of an idea that you have had or scenario you want to explore prior to them making a significant purchasing design in your assigned theme.

You must come up with an idea (or base yours on one of those listed below) to explore with a reason and then undertake your design space exploration. You need to show (and document) that you can

1. Come up with and elaborate an idea that you can explore in your assigned theme area using the design tools you have made in weeks 1 to 7. The more context aware your study is relative to the case study, the better the potential outcome. You are trying to demonstrate how you would use the tools you have developed to explore ideas related to the case study and recommend changes based upon your results.
2. Determine how you can use the design tools to undertake the exploration or design study you have identified in step 1.
3. Use the simulation tools to undertake the study you have outlined in step 2.
4. Present the results from step 3 in an easy to understand and concise form.
5. Summarise the main outcomes and conclusions from step 4
6. Reflect on the outcomes from step 5 including:
    a. what limitations apply to the outcomes of your study(due to tools, data, approach, etc)
    b. how you would overcome these limitations if you had more time
    c. a response to your original design idea from step 1

3.2 Submission Instructions – Part 3

You must submit:

- A report of up to 6 pages that documents the key elements from steps 1 to 6 above
- 1 simulation file that represents the key elements of your study.

Present your work clearly and concisely and remember to use the assessment criteria to guide you. High marks will come from correct and insightful application of the skills learned in this unit NOT lengthy submissions.

Submit your work to “Part 3 – Further Study” in the “Assessments” area of the Blackboard site.

3.3 Example Studies:

	Example 1	Example 2	Example 3	Example 4
Machines and Drivetrain	Impact of the counterweight mass on efficiency and/or total energy usage.	Closed loop control of the machine to improve the current response in simulation	Impact of changing machine pole number alongside gear ratio of efficiency and/or DC Link suitability	Impact of changing operation to a non-salient machine and adopting a maximum torque per amp strategy on efficiency.
Converter	Impact of deadtime on harmonics and motor losses	Impact of DC-link ripple on output harmonics and motor losses	Impact of modulation schemes (not just those covered in activity sheet 7) on harmonics & motor losses	